

Manufacturing Service & Operations Project 2026

Aggregate Planning for Pumphouse Operations

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Workforce & capacity planning for pump-house operations and
maintenance at Aster's Pulau Bukom refinery



Aster

Company Overview



BACKGROUND

Leading Provider of
Chemical & Energy
Solutions in Singapore
and Asia Pacific Region

FOCUS

Provide the energy and
chemical needs of society
through Sustainable
Transformations

OPERATIONS

A refinery with a production capacity
of 300,000 barrels per day and an
ethylene cracker producing 1.1 million
tonnes annually.

Aster Bukom now stands as a world-class, fully integrated oil and chemicals manufacturing hub, which provides transport fuels, base oils, bitumen, base chemicals, and speciality chemicals such as butadiene to customers in Asia and beyond.

Alongside Aster Jurong Island, Aster Bukom supplies energy and advanced chemical products regionally via pipeline links to Jurong Island and Singapore's industrial network.

PULAU BUKOM



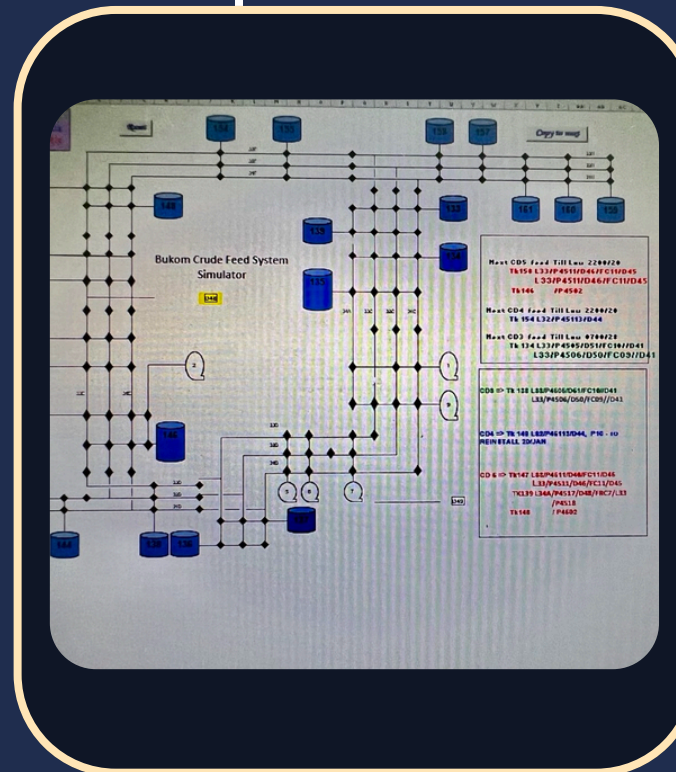
Project Overview

Aggregate Planning of Pumphouse PH43

01

Context

PH43 is a pumphouse unit in Pulau Bukom Refinery that supports refinery fluid transfer operations, especially naphtha-related movements between tanks, process units, and other refinery areas.



02

Planning Problem

PH43 does not produce finished goods directly. Its workload comes from operational support activities such as fluid transfer, tank preparation, pump monitoring, preventive maintenance, corrective maintenance, and shutdown support.

Project Overview

Aggregate Planning of Pumphouse PH43

03

Project Objective

This project develops a aggregate planning model to estimate monthly PH43 workload and determine the manpower capacity required to meet demand.



04

Planning Focus

- Regular workforce capacity
- Overtime usage
- Contractor support
- Backlog control
- Preventive maintenance planning
- Spare-part readiness

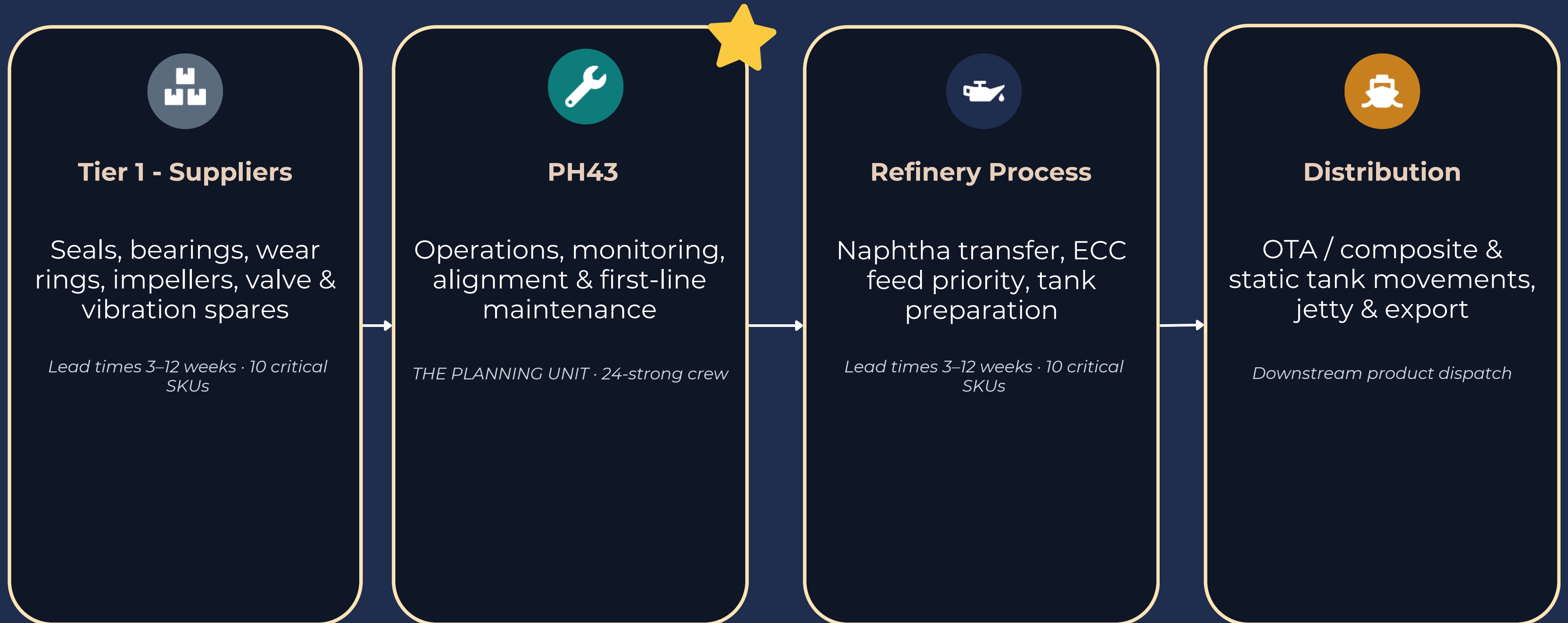
05

Outcome

PH43 Pumphouse Operations
↓
Monthly Workload Drivers
Transfer volume | Tank prep |
Maintenance | Shutdown support
↓
Converted into Crew-Hours
↓
Aggregate Planning Model
↓
Workforce + Overtime + Contractors
+ Backlog + Cost

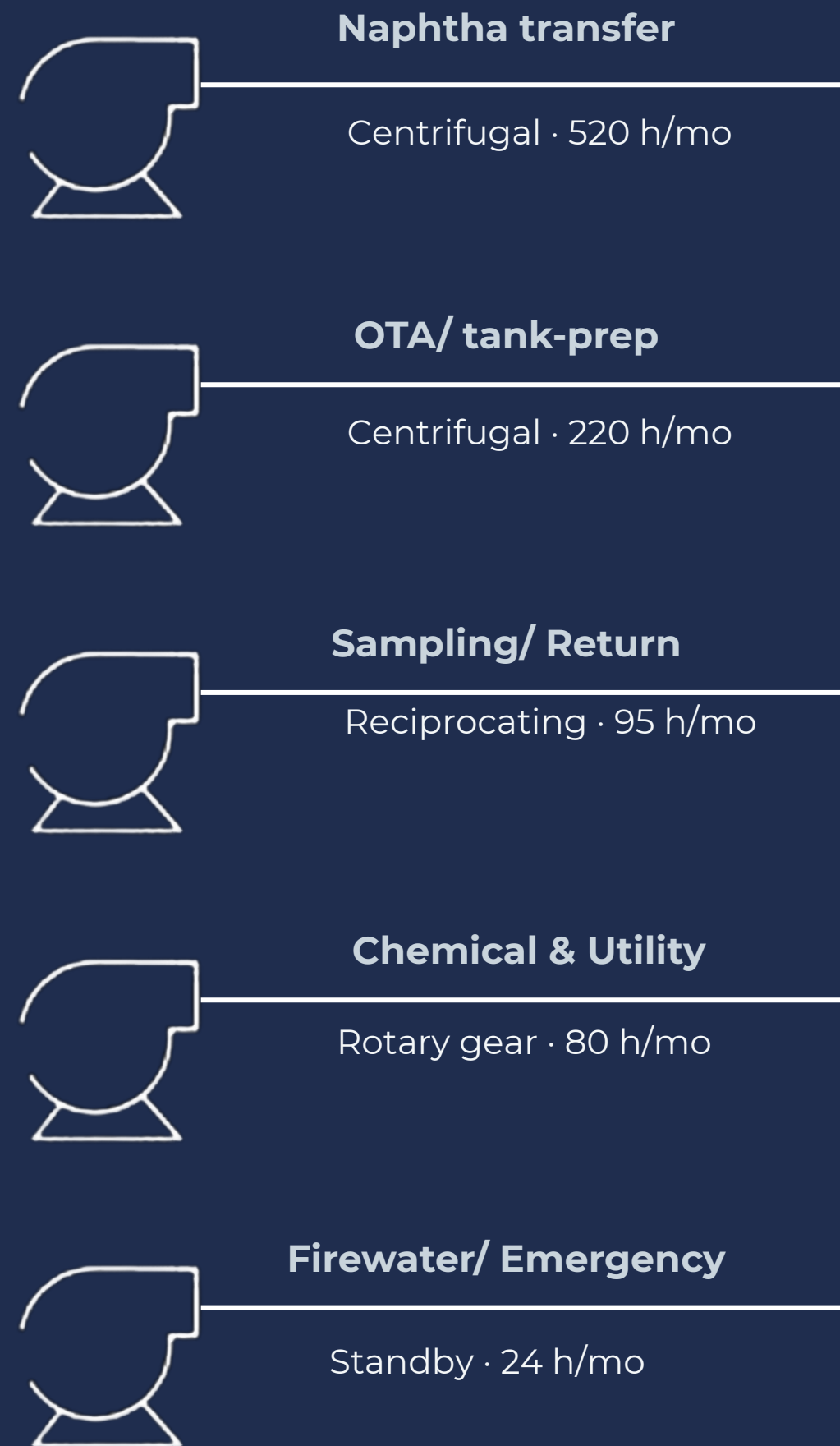
Supply Chain Map

Where does this pumphouse lies in Aster's flow of Material & Products?



Current Approach

Approach	Current Practice	Maturity		Explanation
Demand Forecasting	Demand expressed monthly via transfer volume, PM/CM work orders & shutdown days	Medium		Transfer and maintenance needs are estimated, but demand may fluctuate due to refinery operations.
Capacity Planning	Fixed shift roster of ~24 staff; capacity treated as broadly constant	Low-Medium		Manpower is likely planned around operational needs, but may not be fully optimized month by month.
Maintenance Planning	Maintenance scheduled largely by calendar PM, not levelled against transfer load	Medium		Preventive maintenance is applied.
Cost-Based Planning	Peaks absorbed reactively through overtime and contractor call-off	Low		Overtime, contractor usage, backlog, and spare-part costs are not be compared systematically.



SPHE

Standard Pump-Hour
Equivalent

→ crew-hours

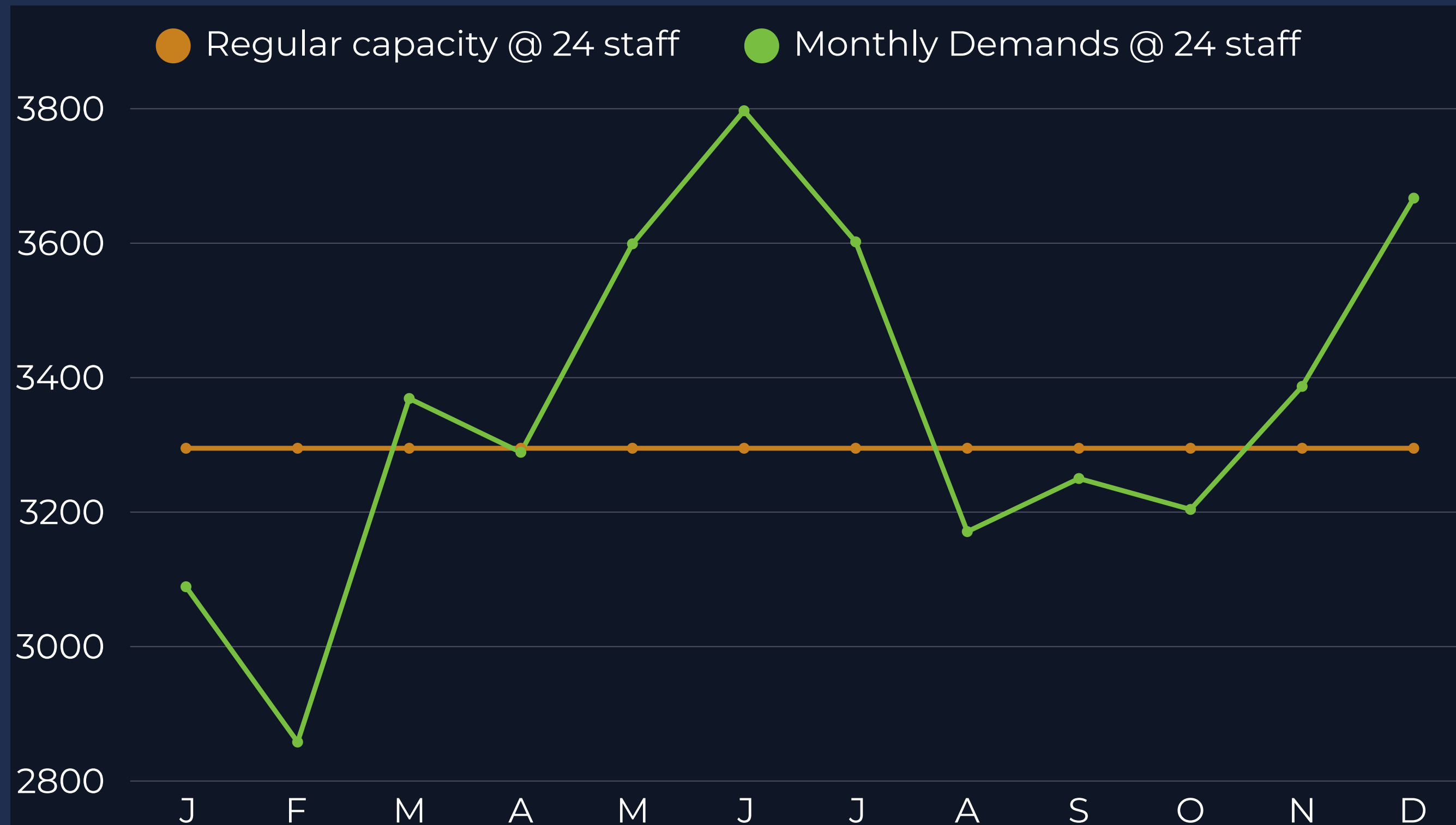
Pros

- Centrifugal · 220 h/Single demand series — one workforce decision
- Pools variability; smooths roster planning
- Common cost basis (SGD/crew-hr)

Cons

- Hides skill mix (operator vs instrument tech)
- Equal weighting can mask safety-critical duties
- Needs disaggregation back to MPS/MRP later

Demand outgrows capacity by mid year



6 out of **12 months**
have demand
exceeding capacity

Average crew hours
per month: **3357**

Peak demand: **3797**

Problem Formulation

12-Month Cost Minimizing Plan

Minimise $\sum_t (5216.64 \cdot W_t + 58 \cdot O_t + 96.32 \cdot C_t + 4200 \cdot H_t + 1800 \cdot L_t + 20.16 \cdot I_t + 128.80 \cdot B_t)$ for $t = 1 \dots 12$



Decision variables

- W_t — workforce headcount (18–28)
- H_t, L_t — workers hired / laid off
- O_t — overtime crew-hours
- I_t — pre-build inventory carried (SPHE)
- C_t — contractor output (SPHE)
- B_t — backlog carried (SPHE)
- (Internal_t = in-house production, SPHE — bounded by capacity, not separately priced)



Constraints

- $W_t = W_{t-1} + H_t - L_t$ (workforce balance)
- $\text{Internal}_t \times 1.12 \leq 137.28 \cdot W_t + O_t$ (capacity: reg + OT)
- $O_t \leq 24.96 \cdot W_t$ (overtime cap)
- $I_{t-1} - B_{t-1} + \text{Internal}_t + C_t - D_t = I_t - B_t$ (supply–demand balance)
- $C_t \leq 580$; $B_t \leq 107$ (contractor & backlog caps, SPHE)
- $18 \leq W_t \leq 28$; $B_{12} = 0$ (bounds; clear backlog by year-end)



Given data

- $D_t = 2,552\text{--}3,390$ SPHE/month (2026 forecast)
- $W_0 = 24$; $176 \text{ h} \times 0.78$ availability = 137.28 productive hrs/worker
- 1 SPHE = 1.12 crew-hours
- Costs from Cost_Inputs sheet (salary basis)
- Service target 0.97 ; calendar-2026 operating days

Discussion

What is the Hybrid Plan



Defensible capacity policy

The hybrid plan keeps the 24-person core crew as the base, while using overtime, work smoothing, and temporary support to handle peak months.



Level remains a strong baseline

Since Level is only around 1% above the LP benchmark under the base costing model, a stable 24-person crew is a practical starting point.



Rolling-horizon control

Re-plan monthly using updated work orders and demand forecasts. Permanent hiring is only considered if overload persists across multiple planning cycles.



Service and fatigue protected

The hybrid plan avoids excessive reactive overtime by shifting movable work to low-load months and reserving overtime or temporary support for unavoidable peak demand.

Key Takeaway: The hybrid plan preserves the stability of Level planning while using LP insights to identify when and where flexible capacity is needed.

Cost comparisons

Strategy	Cost (SGD)	\$/SPHE	Δ vs Opt
LP-optimal	\$1,684,991	46.85	—
Chase	\$1,731,333	48.14	+2.75%
Level	\$1,709,148	47.52	+1.43%
Hybrid	\$1,700,947	46.85	+0.95%

Cost breakdown

Approach	Key Differences					
Chase	60.6k spent on hiring and firing	<table><tr><td>\$46,200</td><td>Hiring</td></tr><tr><td>\$14,400</td><td>Firing</td></tr></table>	\$46,200	Hiring	\$14,400	Firing
\$46,200	Hiring					
\$14,400	Firing					
Level	95.6k spent on overtime	<table><tr><td>\$95,797</td><td>Overtime</td></tr></table>	\$95,797	Overtime		
\$95,797	Overtime					
Hybrid	Trades \$21.5k of cheap pre-build holding to kill \$30k of that overtime	<table><tr><td>\$66,046</td><td>Overtime</td></tr><tr><td>\$21,551</td><td>PM Pre-build (Inventory)</td></tr></table>	\$66,046	Overtime	\$21,551	PM Pre-build (Inventory)
\$66,046	Overtime					
\$21,551	PM Pre-build (Inventory)					

No outsourcing used as in-house capacity (regular + overtime) fully covers peak load

Why not LP Optimal?

Model delta from optimal			
\$1,684,991	+0.95%	+1.43%	+2.75%
LP-optimal cost	Hybrid vs Optimal	Level vs Optimal	Chase vs Optimal

- The costs are derived from calculations that account for monthly manpower hours used which are not reflective of actual salaried employees as they are paid on a monthly basis regardless of the productive manpower hours in a given month
- The LP optimal model suggests an increase and decrease in manpower that mimics a chase strategy for some portions which is not realistic given the time to train new employees, hit to employer reputation and lack of operational stability for a given department
- The LP optimal minimizes total cost but does not prioritize balanced workforce utilization. As a result, periods of increased staffing may still require overtime which reduces the operational benefits of workforce expansion
- Given that the hybrid model is within 1 percent of the LP optimal, we have chosen to use that as it best regulates cost and operational reliability for the planning horizon which reflects better in real world scenarios

Recommendations

Aggregation Schemes, Factors and Rolling Horizons

1 Aggregation

Keep the single SPHE unit for the crew decision, but carry a skill-mix split (operations vs instrument/mechanical) so the plan disaggregates cleanly to MPS/MRP

2 Factors to weigh

Lead times (up to 12 wks for impellers), shutdown calendar, ECC feed-priority spikes and fatigue ceiling on overtime

3 Hybrid capacity policy

The 24 person workforce should be maintained unless sustained increase in demand is seen, boosting operational readiness while the gaps can be filled using overtime and pre loading as it is cheaper than outsourcing

4 Rolling horizon

Re-plan every month over a rolling 12-month window i.e. freeze month 1, keep months 2–12 as guidance and update with fresh work-order forecasts

5 Anticipated benefit

A more realistic and flexible capacity plan that reduces reactive overtime, improves use of low-load months, limits unnecessary permanent headcount and provides a quantified basis for future staffing decisions.

Annex - LP

simplex LP																
Time Period	Inventory (SPHE)	Workforce	Employees Hired	Employees Fired	Overtime (crew-hrs)	Backlog (SPHE)	Internal Production (SPHE)	Contractor (SPHE)		Monthly Demand (SPHE)	Maximum Production (SPHE)	Maximum Overtime (crew-hrs)	balance	Annual Costs by Category		
1	0	22.50	0.00	1.50	0	0	2,758	0	≥	2,758	2,758	562	0	\$1,512,178	Regular Labour	
2	206	22.50	0.00	0.00	0	0	2,758	0	≥	2,552	2,758	562	0	\$28,290	Overtime	
3	0	22.86	0.36	0.00	0	0	2,802	0	≥	3,008	2,802	571	0	\$20,799	Hiring	
4	0	23.96	1.10	0.00	0	0	2,937	0	≥	2,937	2,937	598	0	\$7,786	Firing	
5	0	26.21	2.25	0.00	0	0	3,213	0	≥	3,213	3,213	654	0	\$4,980	PM Pre-build (Inventory)	
6	0	26.21	0.00	0.00	198	0	3,390	0	≥	3,390	3,390	654	0	\$0	Backlog (Stockout)	
7	0	26.21	0.00	0.00	3	0	3,216	0	≥	3,216	3,216	654	0	\$0	Contractor (Outsourcing)	
8	36	23.39	0.00	2.83	0	0	2,867	0	≥	2,831	2,867	584	0	\$1,574,033	Total Variable Costs	
9	0	23.39	0.00	0.00	0	0	2,867	0	≥	2,902	2,867	584	0	\$110,958	Materials (fixed)	
10	6	23.39	0.00	0.00	0	0	2,867	0	≥	2,861	2,867	584	0	\$1,684,991	GRAND TOTAL (incl. materials)	
11	0	24.63	1.24	0.00	0	0	3,019	0	≥	3,024	3,019	615	0	\$46.85	Avg cost (\$/SPHE)	
12	0	24.63	0.00	0.00	286	0	3,274	0	≥	3,274	3,274	615	0			
sum	247		5	4	488	0	35,966	0		35,966	35,966	7,235				

Annex - Chase

chase																
Time Period	Inventory (SPHE)	Workforce	Employees Hired	Employees Fired	Overtime (crew-hrs)	Backlog (SPHE)	Internal Production (SPHE)	Contractor (SPHE)		Monthly Demand (SPHE)	Maximum Production (SPHE)	Maximum Overtime (crew-hrs)	target head count	Annual Costs by Category		
1	0	23.00	0.00	1.00	0	0	2,758	0	≥	2,758	2,819	574	23	\$1,559,775	Regular Labour	
2	0	21.00	0.00	2.00	0	0	2,552	0	≥	2,552	2,574	524	21	\$0	Overtime	
3	0	25.00	4.00	0.00	0	0	3,008	0	≥	3,008	3,064	624	25	\$46,200	Hiring	
4	0	24.00	0.00	1.00	0	0	2,937	0	≥	2,937	2,942	599	24	\$14,400	Firing	
5	0	27.00	3.00	0.00	0	0	3,213	0	≥	3,213	3,309	674	27	\$0	PM Pre-build (Inventory)	
6	0	28.00	1.00	0.00	0	0	3,390	0	≥	3,390	3,432	699	28	\$0	Backlog (Stockout)	
7	0	27.00	0.00	1.00	0	0	3,216	0	≥	3,216	3,309	674	27	\$0	Contractor (Outsourcing)	
8	0	24.00	0.00	3.00	0	0	2,831	0	≥	2,831	2,942	599	24	\$1,620,375	Total Variable Costs	
9	0	24.00	0.00	0.00	0	0	2,902	0	≥	2,902	2,942	599	24	\$110,958	Materials (fixed)	
10	0	24.00	0.00	0.00	0	0	2,861	0	≥	2,861	2,942	599	24	\$1,731,333	GRAND TOTAL (incl. materials)	
11	0	25.00	1.00	0.00	0	0	3,024	0	≥	3,024	3,064	624	25	\$48.14	Avg cost (\$/SPHE)	
12	0	27.00	2.00	0.00	0	0	3,274	0	≥	3,274	3,309	674	27			
sum	0		11	8	0	0	35,966	0		35,966	36,649	7,463				

Annex - Level

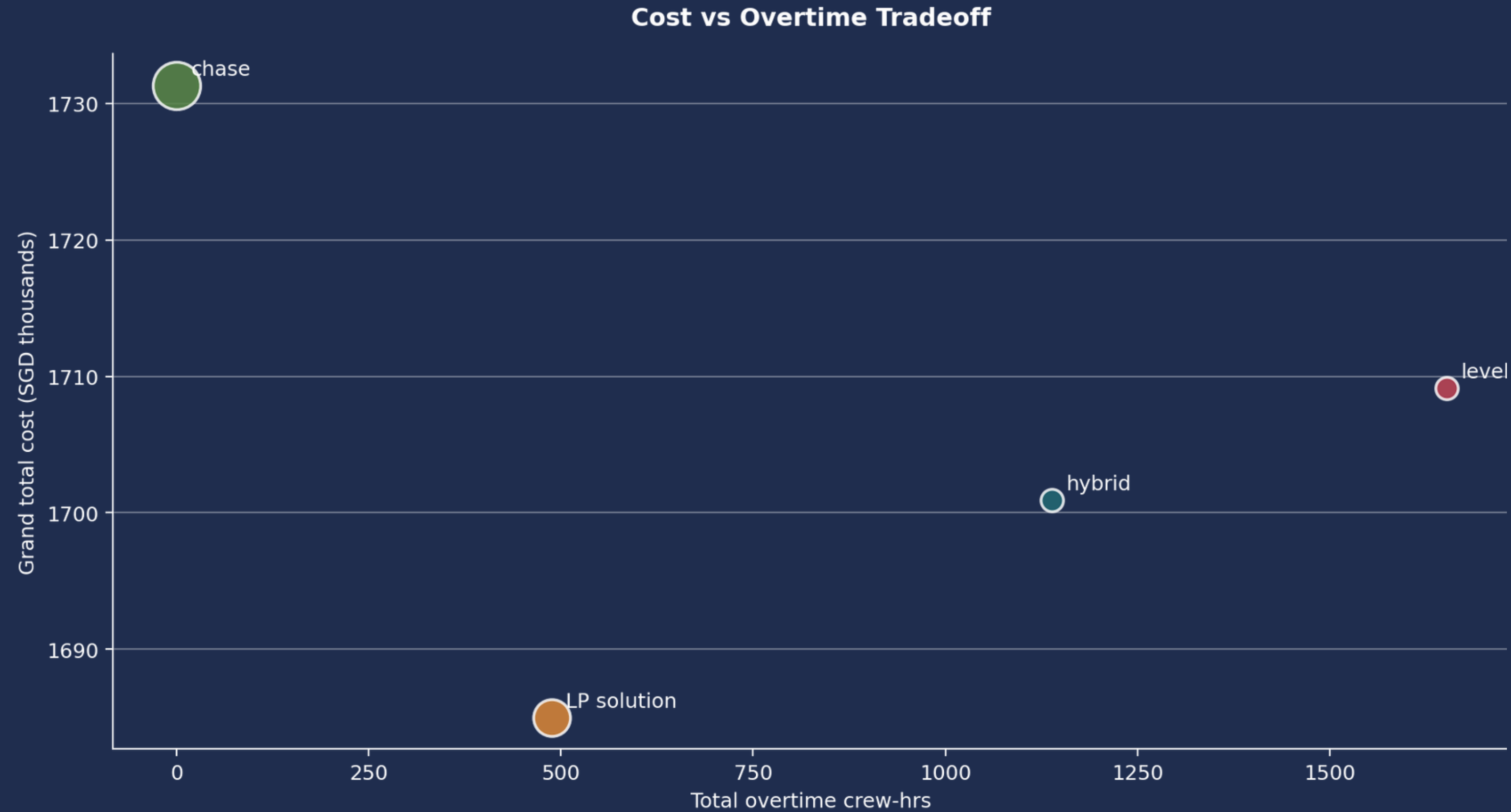
													headcount by principle		
level													24		
Time Period	Inventory (SPHE)	Workforce	Employees Hired	Employees Fired	Overtime (crew-hrs)	Backlog (SPHE)	Internal Production (SPHE)	Contractor (SPHE)		Monthly Demand (SPHE)	Maximum Production (SPHE)	Maximum Overtime (crew-hrs)	balance	Annual Costs by Category	
1	0	24.00	0.00	0.00	0	0	2,758	0	≥	2,758	2,942	599	0	\$1,502,392	Regular Labour
2	0	24.00	0.00	0.00	0	0	2,552	0	≥	2,552	2,942	599	0	\$95,797	Overtime
3	0	24.00	0.00	0.00	74	0	3,008	0	≥	3,008	3,008	599	0	\$0	Hiring
4	0	24.00	0.00	0.00	0	0	2,937	0	≥	2,937	2,942	599	0	\$0	Firing
5	0	24.00	0.00	0.00	304	0	3,213	0	≥	3,213	3,213	599	0	\$0	PM Pre-build (Inventory)
6	0	24.00	0.00	0.00	502	0	3,390	0	≥	3,390	3,390	599	0	\$0	Backlog (Stockout)
7	0	24.00	0.00	0.00	307	0	3,216	0	≥	3,216	3,216	599	0	\$0	Contractor (Outsourcing)
8	0	24.00	0.00	0.00	0	0	2,831	0	≥	2,831	2,942	599	0	\$1,598,190	Total Variable Costs
9	0	24.00	0.00	0.00	0	0	2,902	0	≥	2,902	2,942	599	0	\$110,958	Materials (fixed)
10	0	24.00	0.00	0.00	0	0	2,861	0	≥	2,861	2,942	599	0	\$1,709,148	GRAND TOTAL (incl. materials)
11	0	24.00	0.00	0.00	92	0	3,024	0	≥	3,024	3,024	599	0	\$47.52	Avg cost (\$/SPHE)
12	0	24.00	0.00	0.00	372	0	3,274	0	≥	3,274	3,274	599	0		
sum	0		0	0	1,652	0	35,966	0		35,966	36,775	7,188	0		

Annex - Hybrid

hybrid- workforce stays 24

Time Period	Inventory (SPHE)	Workforce	Employees Hired	Employees Fired	Overtime (crew-hrs)	Backlog (SPHE)	Internal Production (SPHE)	Contractor (SPHE)		Monthly Demand (SPHE)	Maximum Production (SPHE)	Maximum Overtime (crew-hrs)	balance	Annual Costs by Category		
1	0	24.00	0.00	0.00	0	0	2,758	0	≥	2,758	2,942	599		0	\$1,502,392	Regular Labour
2	333	24.00	0.00	0.00	0	0	2,885	0	≥	2,552	2,942	599		0	\$66,046	Overtime
3	267	24.00	0.00	0.00	0	0	2,942	0	≥	3,008	2,942	599		0	\$0	Hiring
4	271	24.00	0.00	0.00	0	0	2,942	0	≥	2,937	2,942	599		0	\$0	Firing
5	0	24.00	0.00	0.00	0	0	2,942	0	≥	3,213	2,942	599		0	\$21,551	PM Pre-build (Inventory)
6	0	24.00	0.00	0.00	502	0	3,390	0	≥	3,390	3,390	599		0	\$0	Backlog (Stockout)
7	0	24.00	0.00	0.00	307	0	3,216	0	≥	3,216	3,216	599		0	\$0	Contractor (Outsourcing)
8	0	24.00	0.00	0.00	0	0	2,831	0	≥	2,831	2,942	599		0	\$1,589,989	Total Variable Costs
9	40	24.00	0.00	0.00	0	0	2,942	0	≥	2,902	2,942	599		0	\$110,958	Materials (fixed)
10	120	24.00	0.00	0.00	0	0	2,942	0	≥	2,861	2,942	599		0	\$1,700,947	GRAND TOTAL (incl. mate
11	38	24.00	0.00	0.00	0	0	2,942	0	≥	3,024	2,942	599		0	\$47.29	Avg cost (\$/SPHE)
12	0	24.00	0.00	0.00	329	0	3,236	0	≥	3,274	3,236	599		0		
sum	1,069		0	0	1,139	0	35,966	0		35,966	36,317	7,188				

Annex - Cost vs Overtime



Bubble size represents workforce churn: hires plus fires.

Annex - Cost Ladder

Contractor was offered to the model as a costed surge lever and selected at zero in every month, confirming in-house capacity (regular + overtime) fully covers peak load.

Source	Cost / SPHE	
Regular time	~\$42.6	salary capacity already paid for
Pre-build	+\$20.16/mo held	cheap, uses idle regular capacity
Overtime	\$64.96	premium, but still in-house
Contractor	\$96.32	most expensive